National University of Computer and Emerging Sciences
Lahore Campus

Digital Logic Design (EE1005)

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 45
Total Questions: 3

Date: 21st February, 2026

Course Instructor(s)
Dr. Muhammad Naeem
Ms. Tamania Javaid

Roll No ____ Section ____ Student Signature ____

Instructions:

- Attempt all the questions.
- Show complete working of each question.
- Multiple solutions of the same question will carry zero credit.
- State your valid assumptions clearly if you have to take any.

CLO #1: Describe various number systems and perform arithmetic operations and base conversions

[15 marks]

Q1:

Convert the following numbers from the given base to the other three bases listed in the table. Show complete working on the answer sheet.

Decimal	Binary	Octal	Hexadecimal
69.3125	?	?	?
?	11101.0011	?	?
?	?	56.74	?
?	?	?	3B.02A

CLO #2: Apply Boolean algebra and K-map methods to optimize logic circuits

[15 marks]

Q2:

Consider the following Boolean expression

$$F(A,B,C,D) = (\bar{A} + \bar{B} + D)(\bar{A} + \bar{D})(A + B + \bar{D})(A + \bar{B} + C + D)$$

a) **Apply** the K- map method to find the optimized Boolean expression of the system in POS (product of sum) form.

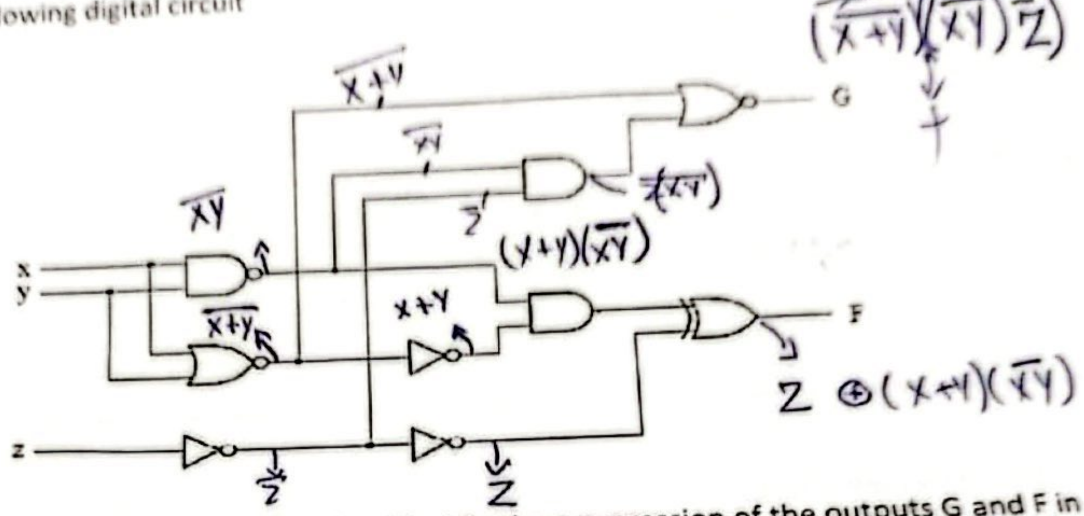
b) **Construct** the logic circuit from the simplified expression using 2 input NAND gates only.

$(B+\bar{D})(\bar{A}+\bar{B})(\bar{B}+C+D)$

CLO #2: Apply Boolean algebra and K-map methods to optimize logic circuits

[15 marks]

Q3:
Consider the following digital circuit



Construct the truth table and **obtain** the simplified Boolean expression of the outputs G and F in *terms of inputs X, Y and Z.*

$Z \overline{XY} \mid X + Y -$

NAND

NOR

XOR

$(\bar{A}B + A\bar{B})$

$(A \oplus B)$